DECLARATION BY THE CANDIDATE

We the undersigned solemnly declare that the report of the project work entitled **Detecting Realtime Sentiment Gaps Across Social Media Ecosystems** is based on our own work carried out during the course of my study under the supervision of **Prof. Revati Raman Devangan.**

I assert that the statements made and conclusions drawn are an outcome of the project work. I further declare that to the best of my knowledge and belief that the report does not contain any part of any work which has been submitted for the award of any other degree/diploma/certificate in this University/ any other University of India or any other country.

(Signature of Student)	**(Signature of Student)**	**(Signature of Student)**
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APPROVAL CERTIFICATE

This is to Certify that the report of the project submitted is an outcome of the project work entitled
Detecting Realtime Sentiment Gaps Across Social Media Ecosystems

carried out by **Rajat Yadaw bearing Roll No.:300102223084, Enrollment No.:CD8356; Parth Hiwarkar bearing Roll No.:300102223074, Enrollment No.:CD8398 ; Virendra Mandavi bearing Roll No.:300102223118, Enrollment No.:CD8337**

Under my guidance and supervision in partial fulfillment of Bachelor of Technology in Computer Science from Bhilai Institute of Technology, Durg, an institute affiliated to Chhattisgarh Swami Vivekanand Technical University, Bhilai (C.G).

To the best of my knowledge and belief the project

i) Embodies the work of the candidate himself / herself,
ii) Has duly been completed,
iii) Fulfills the requirement of the Ordinance relating to the B. Tech. degree of the University,
iv) Is up to the desired standard for the purpose of which is submitted.

(Signature of the Supervisor)

Prof. Revati Raman Devangan

Computer Science Engineering.

The Project work as mentioned above is hereby being recommended and forwarded for examination and evaluation.

Dr. (Mrs,) Sunita Soni Head of the Department

Computer Science & Engineering

CERTIFICATE BY THE EXAMINERS

This is to Certify that the project the entitled

Detecting Realtime Sentiment Gaps Across Social Media Ecosystems

Submitted by

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We have great pleasure in the submission of this project report entitled **“Detecting Realtime Sentiment Gaps Across Social Media Ecosystems”** in partial fulfilment of the degree of Bachelor of Technology.

While submitting this project report, we take this opportunity to thank those directly or indirectly related to the project work.

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ABSTRACT

Social media platforms have become one of the primary sources for expressing opinions, sharing experiences, and discussing current events. Every day, millions of users generate large volumes of textual content across platforms such as YouTube, Reddit, Hacker News, Dev.to, and other online communities. Analysing this data manually is challenging due to its volume, diversity, and dynamic nature. Consequently, there is a growing need for intelligent systems capable of automatically extracting insights and understanding public sentiment from social media content.

The project “Detecting Realtime Sentiment Gaps Across Social Media Ecosystems” presents an AI-driven solution for collecting, analysing, and comparing sentiments expressed across multiple social media platforms. The system utilizes Natural Language Processing (NLP) and Deep Learning techniques to classify user comments into Positive, Negative, and Neutral sentiment categories. A custom Bidirectional Long Short-Term Memory (BiLSTM) network integrated with an Attention mechanism is employed to improve sentiment classification accuracy by capturing contextual relationships within textual data.

The proposed system gathers comments and discussions from various social media sources, preprocesses the collected text, performs sentiment analysis, and visualizes the results through an interactive dashboard. The dashboard provides platform-wise sentiment distribution, sentiment rankings, heatmaps, classified comment samples, and sentiment gap analysis, enabling users to identify differences in public opinion across platforms. Additionally, the system incorporates real-time YouTube comment and live-chat monitoring along with negative sentiment spike detection to identify sudden changes in audience reactions.

The developed solution demonstrates how Artificial Intelligence can be leveraged to transform large-scale unstructured social media data into meaningful insights. The project has potential applications in brand monitoring, market research, public opinion analysis, reputation management, political campaigns, product feedback evaluation, and trend analysis. By enabling real-time cross-platform sentiment comparison, the system provides a comprehensive understanding of online public perception and emerging sentiment trends.

Keywords: Sentiment Analysis, Natural Language Processing, Deep Learning, BiLSTM, Attention Mechanism, Social Media Analytics, Real-Time Monitoring, Sentiment Gap Detection, Data Visualization, Artificial Intelligence.

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List of Abbreviation

Acronym	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
Bi-LSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
DL	Deep Learning
EDA	Exploratory Data Analysis
GUI	Graphical User Interface
HTML	HyperText Markup Language
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
LSTM	Long Short-Term Memory
ML	Machine Learning
NLP	Natural Language Processing
NLTK	Natural Language Toolkit
OOV	Out Of Vocabulary
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
SGD	Stochastic Gradient Descent
SVM	Support Vector Machine
UI/UX	User Interface / User Experience
URL	Uniform Resource Locator